

Zolotarev 1a

A.C. Antoulas¹, I-V. Gosea², and C. Poussot-Vassal^{*3}

¹Department of Electrical and Computer Engineering, Rice University, Houston
(aca@rice.edu)

²Max Planck Institute, CSC Group, Magdeburg, Germany
(gosea@mpi-magdeburg.mpg.de)

³ONERA, DTIS, Université de Toulouse, France
(charles.poussot-vassal@onera.fr)

June 26, 2026

Abstract

Zolotarev

Contents

1	Detailed examples exposition and results	2
2	$r = 3$	3
2.1	Numerator, denominator, poles and zeros	3
2.2	Plots	3
3	$r = 4$	5
3.1	Numerator, denominator, poles and zeros	5
3.2	Plots	5

^{*}Corresponding author: charles.poussot-vassal@onera.fr

1 Detailed examples exposition and results

2 $r = 3$

2.1 Numerator, denominator, poles and zeros

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$1.8 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-6.8 \cdot 10^{-15}$	$-1.2 \cdot 10^{-15}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$1.9 \cdot 10^{-16}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-4.2 \cdot 10^{-15}$	$-1.3 \cdot 10^{-15}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$4.4 \cdot 10^{-16}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-2.8 \cdot 10^{-15}$	$-2.0 \cdot 10^{-16}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 1: Case 1a, $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	0	$6.58 \cdot 10^{-16} + 4.81 \cdot 10^{-17}i$	$-0.00323 - 9.0 \cdot 10^{-5}i$	$-0.00181 + 0.493i$
2.0	$1.5i$	$2.14 \cdot 10^{-15} - 2.06i$	$0.00557 + 1.5i$	$-0.00165 - 0.494i$
3.0	$-1.5i$	$1.49 \cdot 10^{-15} + 2.06i$	$0.00428 - 1.5i$	$180.0 - 31.3i$

Table 2: Case 1a, $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	NaN	NaN	$-0.00241 + 0.5i$	$-0.00236 + 1.2 \cdot 10^{-4}i$
2.0	$0.5i$	$9.48 \cdot 10^{-16} - 0.637i$	$-0.00245 - 0.501i$	$0.00667 - 1.48i$
3.0	$-0.5i$	$8.89 \cdot 10^{-16} + 0.637i$	$237.0 + 13.9i$	$0.00871 + 1.48i$

Table 3: Case 1a, $r = 3$, Z4: zeros.

2.2 Plots

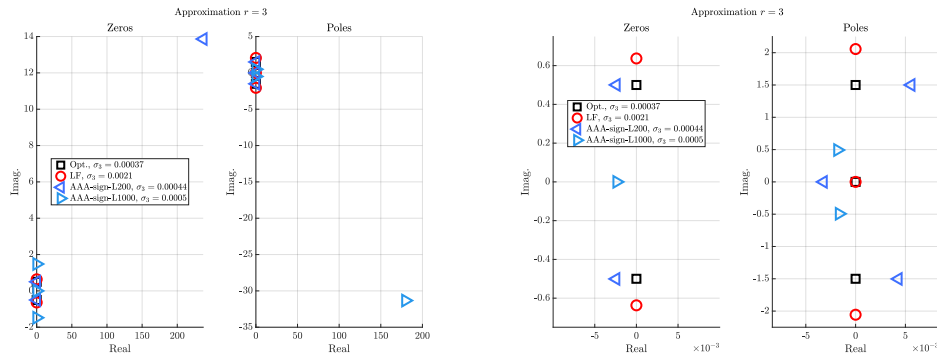


Figure 1: Case '1a', $r = 3$. Left: zeros (left) and poles (right). Right: discarding the large values.

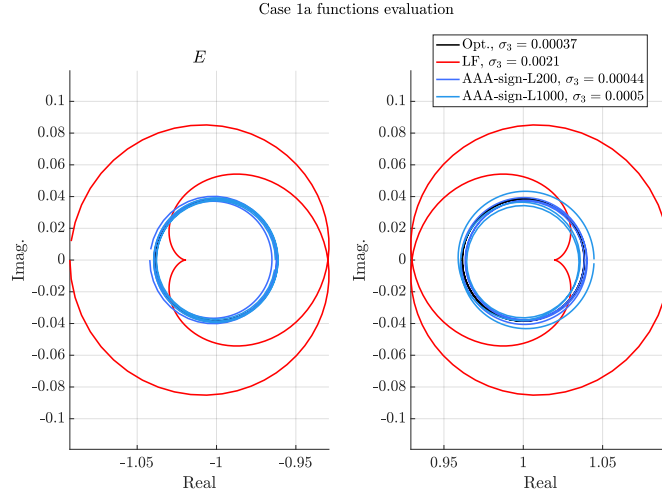


Figure 2: Case '1a', $r = 3$. E and F evaluations.

3 $r = 4$

3.1 Numerator, denominator, poles and zeros

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	-140.0	16.0	-150.0	7.3
z^3	3.5	0	3.7	$4.1 \cdot 10^{-16}$	4.7	-0.29	4.8	-0.22
z^2	0	0	$-4.6 \cdot 10^{-15}$	$3.8 \cdot 10^{-16}$	-650.0	70.0	-660.0	33.0
z	2.6	0	3.2	$2.0 \cdot 10^{-16}$	5.5	-0.52	5.6	-0.34
1.0	0	0	$-1.3 \cdot 10^{-15}$	$-1.6 \cdot 10^{-15}$	-80.0	8.8	-82.0	4.1
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-2.7 \cdot 10^{-15}$	$-8.4 \cdot 10^{-16}$	-500.0	54.0	-510.0	25.0
z^2	4.5	0	5.2	$8.2 \cdot 10^{-16}$	7.9	-0.7	7.9	-0.49
z	0	0	$-3.5 \cdot 10^{-15}$	$-1.1 \cdot 10^{-15}$	-370.0	41.0	-380.0	19.0
1.0	0.56	0	0.75	$1.0 \cdot 10^{-16}$	1.4	-0.13	1.4	-0.078

Table 4: Case 1a, $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$3.21 \cdot 10^{-16} - 0.384i$	$0.00377 + 6.65 \cdot 10^{-5}i$	$0.00375 - 2.07 \cdot 10^{-5}i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$3.09 \cdot 10^{-16} + 0.384i$	$0.00535 - 0.864i$	$0.00513 + 0.864i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$1.12 \cdot 10^{-15} - 2.25i$	$0.00516 + 0.864i$	$0.00532 - 0.864i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$9.09 \cdot 10^{-16} + 2.25i$	$499.0 - 54.4i$	$507.0 - 25.2i$

Table 5: Case 1a, $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00403 - 0.358i$	$0.00404 + 0.358i$
2.0	0	$4.1 \cdot 10^{-16} + 5.92 \cdot 10^{-16}i$	$0.00406 + 0.358i$	$0.00402 - 0.358i$
3.0	$0.866i$	$3.69 \cdot 10^{-16} + 0.929i$	$0.0128 - 2.08i$	$0.0127 - 2.09i$
4.0	$-0.866i$	$4.46 \cdot 10^{-16} - 0.929i$	$0.0117 + 2.09i$	$0.0116 + 2.09i$

Table 6: Case 1a, $r = 4$, Z4: zeros.

3.2 Plots

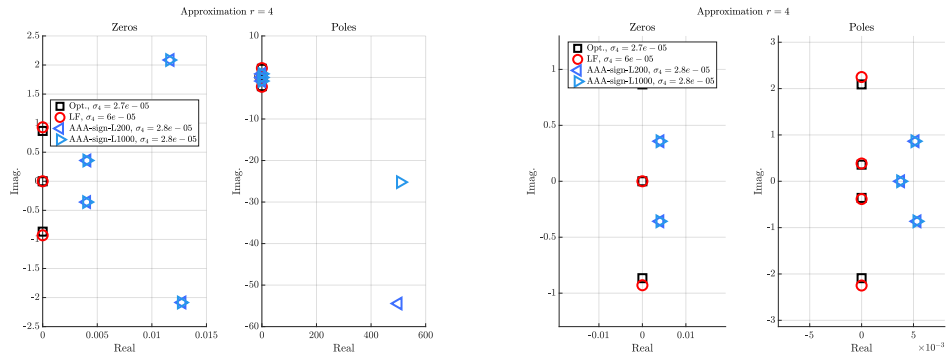


Figure 3: Case '1a', $r = 4$. Left: zeros (left) and poles (right). Right: discarding the large values.

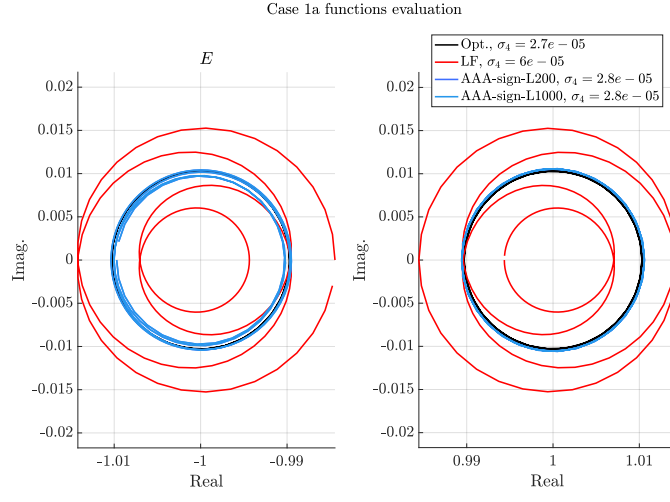


Figure 4: Case '1a', $r = 4$. E and F evaluations.